

4(a)	<table border="1"> <thead> <tr> <th data-bbox="627 218 972 323">statement</th><th data-bbox="972 218 1310 323">genetic engineering using a plasmid</th><th data-bbox="1310 218 1650 323">gene editing</th></tr> </thead> <tbody> <tr> <td data-bbox="627 323 972 459">It may cause the organism to produce a different protein.</td><td data-bbox="972 323 1310 459">✓</td><td data-bbox="1310 323 1650 459">✓ ;</td></tr> <tr> <td data-bbox="627 459 972 595">It may cause a single base pair in a gene to be changed.</td><td data-bbox="972 459 1310 595"></td><td data-bbox="1310 459 1650 595">✓ ;</td></tr> <tr> <td data-bbox="627 595 972 762">The success of the technique can be evaluated using marker genes.</td><td data-bbox="972 595 1310 762">✓</td><td data-bbox="1310 595 1650 762">;</td></tr> <tr> <td data-bbox="627 762 972 866">It may use the CRISPR system.</td><td data-bbox="972 762 1310 866"></td><td data-bbox="1310 762 1650 866">✓ ;</td></tr> <tr> <td data-bbox="627 866 972 978">It uses DNA ligase.</td><td data-bbox="972 866 1310 978">✓</td><td data-bbox="1310 866 1650 978">;</td></tr> </tbody> </table>	statement	genetic engineering using a plasmid	gene editing	It may cause the organism to produce a different protein.	✓	✓ ;	It may cause a single base pair in a gene to be changed.		✓ ;	The success of the technique can be evaluated using marker genes.	✓	;	It may use the CRISPR system.		✓ ;	It uses DNA ligase.	✓	;	5
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4(b)(i)	C ;	1																		

Question	Answer	Marks
4(b)(ii)	any three from: 1 increased / high, (crop) yield ; 2 <i>idea of</i> less food shortages / more food production / helps solve global demand for food ; 3 less need to remove weeds (by hand) ; 4 more income for, growers / farmers or <i>ref. to</i> economic benefit for, country / region ; 5 cheaper / lower cost, to buy ; 6 AVP ; e.g. less need for deforestation of other land	3